

Virtual Technical Conference Platform

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Functional Requirements

REQ ID	TYPE	DESCRIPTION	PRIORITY	ACCEPTANCE CRITERIA	RATIONALE
FR-001	Functional	The system shall allow a Track Chair to schedule a session in their track with a title, speaker, start and end time in UTC and a virtual room, and shall reject any schedule that overlaps an existing booking of the same speaker or the same virtual room.	High	Creating a session that overlaps an existing booking is refused and the conflicting session is named on screen. A non-overlapping session appears on the public agenda within 10s, rendered in each attendee's local time zone.	The agenda drives the whole event; a double-booked speaker or room becomes a visible failure once the conference is live.
FR-002	Functional	The system shall allow a Conference Author to open and host their scheduled session in the virtual meeting room from 15 minutes before the start time, with camera, microphone and screen sharing, and shall optionally record the session and publish it to the on-demand library.	High	The "Start Session" control is enabled at T-15 min and only for the assigned speaker. A shared screen reaches all attendees within 5s of activation. When recording is enabled, the file is listed in the on-demand library within 30 min of the session ending.	Delivering the talk is the core service of the platform; the 15-minute window lets the speaker set up before the audience is admitted.
FR-003	Functional	The system shall allow a registered Attendee to join any live session from the agenda in a single click, and shall refuse entry to unregistered users and to users beyond the licensed concurrent capacity of that session.	High	A registered attendee receives the session audio/video stream within 10s of clicking Join. An unregistered user is redirected to registration and the stream endpoint returns HTTP 403. Once the configured capacity (1000) is reached, the next attendee is placed in a read-only overflow view rather than being disconnected.	Attendance is what the ticket buys, and graceful capacity handling prevents an uncontrolled collapse during a keynote.

FR-004	Functional	The system shall allow an Attendee to post a question of at most 300 characters during a live session and to upvote questions posted by others, keeping the question list ordered by upvote count in real time.	Medium	A posted question is visible to every participant of that session within 2s. Upvoting reorders the list without a page reload. Posting is blocked with a clear message once the Track Chair closes Q&A.	Live Q&A is the interaction that distinguishes attending a session from watching its recording.
FR-005	Functional	The system shall allow a Track Chair to moderate the Q&A of any session in their track by approving, hiding or marking a question as answered, and shall log every moderation action with the moderator identity, question ID and UTC timestamp.	Medium	Hiding a question removes it from attendee view within 2s while it stays visible to the chair. Marking a question answered moves it to the "Answered" tab. Every action appears in the exportable moderation log with actor, action, question ID and timestamp.	A live audience channel needs a code-of-conduct control, and the log makes moderation decisions auditable after the event.

Non-Functional Requirements

REQ ID	TYPE	DESCRIPTION	PRI ORI TY	ACCEPTANCE CRITERIA	RATIONALE
NFR-001	Non-Func	The system shall deliver the live session stream at 720p with end-to-end latency below 5s for 95 % of attendees while 2000 attendees are watching a single session concurrently.	High	A load test with 2000 simulated attendees on one session reports p95 glass-to-glass latency under 5 s, mean bitrate of at least 1.5 Mbps and fewer than 1 % dropped connections over a 60-minute run.	Latency beyond a few seconds breaks the question-and-answer rhythm, and the keynote is the peak load of the whole event.
NFR-002	Non-Func.	The system shall stay available for at least 99.9 % of each scheduled conference day, and shall let a disconnected attendee rejoin a live session and resume playback within 15 s without re-authenticating.	High	Uptime monitoring across a conference day records no more than 1.5 minutes of unavailability. Dropping an attendee's network for 30s results in automatic rejoin and resumed playback within 15s in at least 19 of 20 trials.	A virtual conference runs in a fixed time window, so an outage during a live session cannot be made up afterwards.